DOMAIN 4 STUDY MATERIAL QUESTIONS

Question 1

Question: In the development of emergency response plans, which of the following is the most critical factor that needs to be addressed?

- a. Emergency equipment needed
- b. Emergency personnel
- c. Response to multiple emergencies
- d. How to protect people, property, and the environment

Correct Answer: Select (d)
Explanation: The most important factor to consider is how to protect life, property, and the environment. Although emergency equipment, personnel, and response to multiple emergencies must be addressed in the plan, none of them is the most important factor in the development of the emergency response plan.

Question 2

Question: Strict liability is the absolute legal responsibility for an injury that can be imposed on the wrongdoer without proof of carelessness or fault. Which of the following IS NOT part of what the plaintiff must prove in product liability cases?

- a. That the item was defective
- b. That the defect proximately caused the injury
- c. That the product did not work as expected
- d. That the defect rendered the product unreasonably dangerous

Correct Answer: Select (c)
Explanation: This statement is related to implied warranty. Implied warranty is the assurance that a product will work if it is used for a reasonably expected purpose.

Question 3

Question: Based on a Factory Mutual Engineering Corporation study of 25,000 fires over a period of 10 years, the leading cause of industrial fires is:

- a. Sparks
- b. Smoking
- c. Electrical
- d. Hot surfaces

Correct Answer: Select (c)
Explanation: This study indicated that about 23% of all industrial fires are electrical in nature.

Question 4

Question: The leading cause of civilian fires is:

- a. Electrical
- b. Smoking
- c. Candles
- d. Cooking equipment

Correct Answer: Select (b)
Explanation: An estimated 35% of civilian fires are caused with cigarettes as their ignition source.

Question 5

Question: The standards developed by the National Fire Protection Association (NFPA) for fire prevention and protection are called:

- a. NFPA Code of Fire Standards
- b. Uniform Building Fire Code
- c. Fire Safety Standards
- d. National Fire Code

Correct Answer: Select (d)
Explanation: NFPA standards for fire protection are called National Fire Code. Life Safety Code is an element of the National Fire Code.

The NFPA 101 Life Safety Code is designed with the goal of protecting people in case of a fire in a building. It takes into account the building's construction, occupancy, and protection features. This NFPA code has provisions for means of egress, fire protection features, alarms, sprinkler systems, emergency lighting, and smoke barriers for all types of occupancies.

Question 6

Question: Auto-ignition temperature is the minimum temperature at which a material would burn in the absence of an ignition source. This temperature is:

- a. Lower than ignition temperature
- b. Higher than ignition temperature
- c. The same as ignition temperature
- d. Depending on the material, it could be higher or lower than ignition temperature

Correct Answer: Select (b)
Explanation: Auto-ignition temperature is higher than ignition temperature. This phenomenon occurs as a result of oxidation of fuel which would generate heat and raise the temperature. Higher temperatures would cause higher oxidation rates and even more heat would be generated. This process would continue until the material reaches its auto-ignition temperature. Paint and oil soaked rags have caused many fires in this way. Flammables should never be exposed to temperatures above their auto-ignition temperature.

Question 7

Question: In order for combustion reactions to start and proceed, which of the followings are needed simultaneously?

- a. Fuel, heat, source of ignition, and air
- b. Fuel, spark, oxygen, heat
- c. Fuel, chemical chain reaction, oxygen, source of ignition
- d. Fuel, auto-ignition temperature, air, source of ignition

Correct Answer: Select (c)
Explanation: Fuel, air, and ignition source are obvious elements of combustion. However, we also need chemical chain reaction (formation of free radicals) for the combustion reactions to take place. Actually, halon and dry chemical fire extinguishing agents put out a fire by stopping the chemical chain reactions.

Question 8

Question: Two cylinders of hydrogen and five 55 gallon drums of gasoline burn completely in a storage area made of non-combustible material. Assuming that the density of gasoline is 4 lbs./gallon, its heating value is 20,000 Btu/lb., and that there are 15 lb of hydrogen in each cylinder with a heating value of 55,000 Btu/lb., the amount of heat released is:

- a. $40 \times 10^6\text{ Btu}$
- b. $20 \times 10^5\text{ Btu}$
- c. $23.65 \times 10^6\text{ Btu}$
- d. $32.56 \times 10^6\text{ Btu}$

Correct Answer: Select (c)
Explanation:
$5\text{ drums} \times 55\text{ gal/drum} \times 4\text{ lb/gal} \times 20,000\text{ Btu/lb} = 22 \times 10^6\text{ Btu}$ released from gasoline.
$2\text{ cylinders} \times 15\text{ lb/cylinder} \times 55,000\text{ Btu/lb} = 1,650,000\text{ Btu}$ released from combustion of H_2.
Total Btu released $= 23.65 \times 10^6\text{ Btu}$.

Question 9

Question: A drum of solvent is on fire in a storage room. Assume that heat is transferred from the outer surface of the drum having a temperature of $1,500^\circ\text{F}$ to the wall having a temperature of $1,000^\circ\text{F}$ by convection. If the heat transfer coefficient is $80\text{ Btu/hr/ft}^2/^\circ\text{F}$, the rate of heat transfer is: (Assume a heat transfer area of 50 ft^2.)

- a. $2 \times 10^6\text{ Btu/hr}$
- b. $96,000\text{ Btu/hr}$
- c. $8.9 \times 10^6\text{ Btu/hr}$
- d. $12.7 \times 10^6\text{ Btu/hr}$

Correct Answer: Select (a)
Explanation: Convection is the mechanism of interphase heat transfer, i.e., heat transfer between two different phases. If we have a surface at temperature T_1 that is transferring heat to the surrounding air at a temperature T_2, this heat is transferred by convection because the transfer is between a solid phase and a gaseous phase.

$$Q = HA(T_1 - T_2)$$

Where:
Q is the rate of heat transfer, A is area, T is temperature, H is the heat transfer coefficient.

For the problem at hand:
$$Q = 80(50)(1,500 - 1,000) = 2 \times 10^6\text{ Btu/hr}$$

Question 10

Question: A flammable liquid storage area made of 1 foot-thick brick walls having a thermal conductivity of $0.4\text{ Btu/hr/foot}/^\circ\text{F}$ is on fire. Assuming that heat is transferred from one of the walls having a temperature of 800°F and an area of 100 ft^2 to the other side of the wall, which has a temperature of 200°F, the rate of heat transfer in Btu/hr is:

- a. 80,000
- b. 24,000
- c. 36,000
- d. 8,000

Correct Answer: Select (b)
Explanation: The equation for heat transfer by conduction is:

$$Q = K \frac{A}{L} (T_1 - T_2)$$

where:
Q is the rate of heat transfer
K is thermal conductivity
A is area through which heat is being transferred
T_1, T_2 are temperatures

For the problem at hand:
$$Q = \left(\frac{0.4}{1}\right)(100)(800 - 200) = 24,000\text{ Btu/hr}$$

Question 11

Question: A drum of carbon disulfide (CS_2) is on fire. The combustion products would most likely be:

- a. Sulfur oxides, carbon monoxide, carbon dioxide
- b. Hydrogen sulfide, carbon monoxide
- c. Hydrogen sulfide, water, carbon monoxide, carbon dioxide
- d. Twice as much hydrogen sulfide as CO and CO_2 combined

Correct Answer: Select (a)
Explanation: Most carbon in the fuel is converted to carbon monoxide and carbon dioxide. Sulfur in the fuel is oxidized to mostly sulfur oxides.

Question 12

Question: A water reactive chemical is defined as one that would produce either a flammable gas or a toxic gas when it comes into contact with water or moisture. Which of the following compounds is a water reactive chemical?

- a. Potassium chloride
- b. Potassium

- c. Potassium cyanide
- d. Only (b) and (c) above.

Correct Answer: Select (d)
Explanation: Potassium reacts with water and produces hydrogen gas which is highly flammable. Potassium cyanide reacts with water and generates hydrogen cyanide which is a highly toxic gas.

Question 13

Question: Which of the following is the most important consideration to prevent fire loss in a flammable liquid storage area?

- a. Compatible storage
- b. Use of explosion proof devices
- c. Availability of firefighting equipment
- d. Use of pre-action automatic sprinkler system

Correct Answer: Select (a)
Explanation: Although statements (b), (c) and (d) are important considerations in fire prevention and protection, the most important consideration is compatible storage such as separating ignitors from ignitables.

Question 14

Question: In most fire situations there is not enough oxygen available for conversion of carbon in the fuels to carbon dioxide, and as a result, large amounts of carbon monoxide are produced. CO has been the primary killer in most fires. This gas can cause fatalities because:

- a. It in the most toxic of all gases produced in a fire.
- b. It is more reactive with blood than oxygen.
- c. It reacts with oxygen in the lungs and causes asphyxiation.
- d. Only (b) and (c) above.

Correct Answer: Select (b)
Explanation: Carbon monoxide which is produced during a fire as a result of incomplete combustion of carbon containing materials is about 210 time more reactive with blood than oxygen. Therefore, CO deprives the body of oxygen and causes death by a form of asphyxiation. Inhalation of air containing 0.4% carbon monoxide can cause death in less than one minute.

Question 15

Question: The gaseous products of combustion have a vapor density of less than 1. This means that air in a fire situation is at:

- a. Low spots
- b. High spots
- c. Mixed with other gases
- d. Halfway between floor and ceiling

Correct Answer: Select (a)
Explanation: Vapor density is defined as the ratio of density of a gas or vapor to the density of air at a given temperature. A vapor density of less than 1 indicates that the gas is lighter than air and would rise in air, a vapor density of greater than 1 indicates that the gas is heavier than air and would sink in air. For the problem at hand, fire gases rise in air, and therefore, air would be collected at the low spots.

Question 16

Question: The following data is available for vapor pressure of acetone as a function of temperature.

Temperature ($^\circ\text{F}$)	**130°**	**120°**	**110°**	**100°**	**90°**
Vapor Pressure (mm Hg)	760	740	720	700	695

When atmospheric pressure is 740 mm Hg, the vapor pressure of acetone is also 740 mm Hg at 120°F. This means that the boiling point of acetone at 740 mm Hg is:

- a. Equal to 120°F
- b. Higher than 120°F but less than 130°F
- c. Lower than 120°F but more than 110°F
- d. Vapor pressure is a measure of the volatility of a liquid and has nothing to do with boiling point.

Correct Answer: Select (a)
Explanation: Boiling takes place at the temperature where liquid vapor pressures equals the ambient (atmospheric) pressure. Vapor pressure of a liquid always increases with an increase in temperature.

Question 17

Question: Acetone is a highly flammable liquid (flash point $= 0^\circ\text{F}$) and has a specific gravity of 0.79. From the data presented, and in the absence of other information, which of the following statements is true regarding a mixture of acetone and water?

- a. Acetone would float on top of water.
- b. Acetone would sink in water.
- c. The mixture has a flashpoint of less than 0°F.
- d. Data presented is not sufficient to determine whether acetone would float or sink in water.

Correct Answer: Select (d)
Explanation: Gases have relatively high values of diffusion coefficient. Therefore, vapor density alone would dictate whether a gas would rise or sink in air. In dealing with liquids, however, we need two pieces of information in order to decide whether a liquid would float or sink in water. The information required is specific gravity and solubility.

Specific gravity is the ratio of density of a liquid to the density of water. If two liquids are insoluble in each other, the liquid with the smaller value of specific gravity would float on top of the other liquid. However, when two liquids are soluble in each other, they become dissolved and form a uniform solution regardless of their specific gravities.

Question 18

Question: Water has a vapor pressure of 20 mm Hg at room temperature and acetone has a vapor pressure of 180 mm Hg at room temperature. This indicates:

- a. Water evaporates faster than acetone.
- b. Water evaporates more slowly than acetone.
- c. At a given pressure, acetone has a lower boiling point compared to water.
- d. Only (b) and (c) above.

Correct Answer: Select (d)
Explanation: Vapor pressure is a measure of force exerted by the vapor to the inside surface of its container. Vapor pressure is also a measure of the volatility of a liquid. The higher the vapor pressure, the more volatile the liquid. Boiling takes place when atmospheric pressure and vapor pressure are equal.

Question 19

Question: Fire point is defined as the lowest temperature at which enough vapor is generated to support continuous combustion in presence of an ignition source. Fire point is:

- a. Lower than flash point
- b. Higher than flash point
- c. Same as flash point
- d. Depending on the material can be higher, lower, or the same as flash point

Correct Answer: Select (b)
Explanation: Flash point is the lowest temperature at which a liquid gives off enough vapor to support combustion. The liquid would not continue to burn at its flash point. Fire point temperature requires sustained combustion and is always higher than flash point temperature.

Question 20

Question: Mishandling of flammable liquids and gases have been responsible for approximately 18 per cent of all industrial fires. In order to safely handle these materials, one has to understand their properties. Flammable liquids are incompressible while flammable gases are compressible. A container with a volume of 100 ft^3 is filled with gasoline. The container is under 100 psi pressure. If this container breaks, the volume of gasoline spilled is approximately:

- a. $100,000\text{ ft}^3$
- b. $10,000\text{ ft}^3$
- c. 100 ft^3
- d. 10 ft^3

Correct Answer: Select (c)
Explanation: Liquids are incompressible. This means that pressure changes have very little effect on the liquid volume.

Question 21

Question: A cylinder of hydrogen having a volume of 5 ft^3 is under 1,000 psig pressure. The valve on the cylinder breaks releasing hydrogen to atmosphere. The volume of hydrogen released is approximately:

- a. 5 ft^3
- b. $5,000\text{ ft}^3$

- c. 345 ft^3
- d. 456 ft^3

Correct Answer: Select (c)
Explanation:
$P_1 = 1,000\text{ psig}$
$V_1 = 5\text{ ft}^3$
$P_2 = 14.7\text{ psia}$

$$P_1 V_1 = P_2 V_2$$

$P_1 = 1,000 + 14.7 = 1,014.7\text{ psia}$ (We must use absolute pressure)
$$(1,014.7)(5) = 14.7 V_2$$
$$V_2 = 345.13\text{ ft}^3 \approx 345\text{ ft}^3$$

Note 1: Gases are compressible which means that the volume of gas is a very strong function of pressure. Note that due to pressure reduction the volume increased from 5 ft^3 to 345 ft^3.
Note 2: In using gas laws be sure to use absolute pressure (gauge pressure + atmospheric pressure) and absolute temperature.

Question 22

Question: A Lower Flammability Limit (LFL) of 10% for a gas or vapor means that the gas would support combustion at:

- a. Concentrations with air below 10%
- b. Concentrations above 10% but below the UFL
- c. Concentrations 10% above UFL
- d. Only (b) and (c) above

Correct Answer: Select (b)
Explanation: LFL is defined as the lowest concentration by volume of a gas in air which would support combustion. UFL is the highest concentration of a gas in air (by volume) that would support combustion. A gas or vapor can only support combustion at concentrations with air that fall between its LFL and UFL. It should be pointed out that both LFL and UFL are functions of ambient pressure and temperature.

Question 23

Question: Fifty gallons of gasoline is spilled from a drum. The specific gravity of gasoline is 0.8 and its vapor density is 2.5. The volume of vapor that can be generated from the spill is approximately:

- a. $1,776\text{ ft}^3$
- b. $1,025\text{ ft}^3$
- c. 125 ft^3
- d. 652 ft^3

Correct Answer: Select (a)
Explanation: Vapor volume can be estimated from:

$$V = \frac{(\text{S.G.})(111)}{\text{V.D.}}$$

Where:
V is volume of vapor generated per gallon of liquid vaporized in cubic feet
S.G. is liquid specific gravity
V.D. is vapor density.

$$V = \frac{(0.8)(111)}{2.5} = 35.52\text{ ft}^3\text{/gal}$$
$$35.52\text{ ft}^3\text{/gal} \times 50\text{ gal} = 1,776\text{ ft}^3$$

Question 24

Question: Methyl alcohol has a vapor volume of 80 ft^3 per gallon and a lower flammability limit of 6% by volume in air. A bottle of methyl alcohol containing 0.5 gallon of methyl alcohol liquid breaks in a room with dimensions of $15 \times 15 \times 10\text{ feet}$. The content of the room is:

- a. Above LFL
- b. Flammable
- c. Below LFL
- d. Only (a) and (b) above

Correct Answer: Select (c)
Explanation:
Vapor generated $= 0.5 \times 80 = 40\text{ ft}^3$
Volume of room $= 15 \times 15 \times 10 = 2,250\text{ ft}^3$
Partial volume of methyl alcohol $= \frac{40}{2250 + 40} = 0.0174$
Concentration $= 1.74 \times 10^{-2}$ or $1.74%$ (which is below LFL of 6%).

Question 25

Question: Materials that are classified as oxidizers must be separated from flammables because:

- a. They can produce heat and act as a source of ignition.
- b. If mixed with flammables may cause a fire in the absence of a source of ignition.
- c. They can release oxygen or catalyze the combustion reactions.
- d. Only (b) and (c) above.

Correct Answer: Select (d)
Explanation: Compounds which are classified as oxidizers are those which can either release oxygen or catalyze the combustion reactions (such as chlorine). In either case, oxidizers would make an ongoing fire much worse and must be separated from flammables. They can also react with flammables and raise the temperature to above ignition temperature.

Question 26

Question: Fire resistance rating is a rating for buildings that:

- a. Is based on laboratory tests.
- b. Determines how long a structure withstands a fire.
- c. Has units of minutes or hours.
- d. All of the above.

Correct Answer: Select (d)
Explanation: "Fire resistance rating" is an indication of how long a given structure can withstand a fire to give the occupants a chance to evacuate the building before commencement of fire fighting procedures.

Question 27

Question: A storage area made of concrete stores 20 drums of gasoline each containing 200 lb of gasoline. The heat of combustion for gasoline is $20,000\text{ Btu/lb}$. The fire load for this storage area is:

- a. $4 \times 10^6\text{ Btu}$
- b. $8 \times 10^7\text{ Btu}$
- c. $6 \times 10^7\text{ Btu}$
- d. $2 \times 10^7\text{ Btu}$

Correct Answer: Select (b)
Explanation:
$$20 \times 200 \times 20,000 = 80,000,000 = 8 \times 10^7\text{ Btu}$$
In calculating the fire load, it is assumed that all combustibles will burn and release their heating value.

Question 28

Question: Fixed temperature heat detectors are not suitable for fires that have the potential for developing quickly because:

- a. They can only respond to rate of rise of temperature.
- b. The air reaches a higher temperature before the device.
- c. In a quick fire heat does not transfer to the device properly.
- d. They can only detect heat coming from a fixed temperature.

Correct Answer: Select (b)
Explanation: There is a time lag before the device temperature reaches the temperature of air surrounding it. Due to this limitation these fire detectors are not suitable for fast developing fires.

Question 29

Question: Most fixed temperature heat detectors use:

- a. Fusible metal
- b. Continuous wires separated by insulation
- c. Bimetallic strips or disks
- d. All of the above

Correct Answer: Select (d)
Explanation: Fusible metal is selected to melt at a given temperature. This would activate an alarm or sprinkler system. The insulation in continuous wire detector is selected in such a way to melt at a given temperature. This closes an electrical circuit which could initiate other actions. Bi-metallic strips consist of two metals with different coefficient of thermal expansion. This would cause the metal to expand at different rates resulting in bending of the strip which can initiate other actions.

Question 30

Question: Rate of rise fire detectors:

- a. Are suitable for fast developing fires
- b. Respond to changes in ambient temperature and pressure
- c. Are classified as heat detectors
- d. Only (a) and (c) above

Correct Answer: Select (d)
Explanation: Rate of rise detectors respond to how fast temperature in the surrounding would rise. These detectors are suitable for fires that can develop rather quickly.

Question 31

Question: Which of the following statements best describe the operation of a rate compensation fire detector?

- a. Responds to a fixed temperature
- b. Is more sensitive to thermal lag than a fixed temperature device
- c. Temperatures that rise rapidly are detected sooner
- d. Only (a) and (c) above.

Correct Answer: Select (d)
Explanation: These detectors respond to temperatures that rise rapidly and have exceeded a pre-set fixed temperature.

A rate compensated detector operates when the ambient air temperature reaches a predetermined level, regardless of the rate at which it increases. The detector will operate at its design temperature during slow-building fire conditions or will respond promptly during rapid fire buildup.

Question 32

Question: Ionization fire detectors respond to:

- a. Smoke particles
- b. Flame
- c. Heat
- d. Ions produced during a fire

Correct Answer: Select (a)
Explanation: Ionization detectors ionize the air and make it conductive. Smoke particles decrease this conductivity and initiate a subsequent action such as alarm or activation of sprinkler system.

Question 33

Question: A dry powder fire extinguishing agent can be used on:

- a. Class A, B, C fires
- b. Metal fires
- c. Class C and D fires
- d. All of the above

Correct Answer: Select (b)
Explanation: The term "dry powder" refers to extinguishing agents designed for metal fires (class D). It should be pointed out that a given dry powder is not suitable for all metal fires. Therefore, class D fire extinguishers must have a plate on the extinguisher that specifies what type of metal fires the extinguishing agent is suitable for.

Question 34

Question: Which of the following gases is expected to be produced from use of Halon 1202 on a fire?

- a. HBr, HF
- b. HCl, HBr, HF
- c. HCl, HF
- d. HI, HCl, HBr, HF

Correct Answer: Select (a)
Explanation: Halon 1202 has two atoms of fluorine and 2 atoms of bromine and its use on a fire would most probably produce hydrogen fluoride and hydrogen bromide.

The four-digit rating of Halon fire extinguishers defines the makeup of the molecule as follows:

- The first numeral is the number of carbon atoms
- The second numeral is the number of fluorine atoms
- The third numeral is the number of chlorine atoms
- The fourth numeral is the number of bromine atoms

Therefore, Halon 1202 has one carbon atom, two fluorine atoms, and two bromine atoms. This indicates that Halon 1202 can be made from methane (CH_4, a highly flammable gas) by substituting two hydrogen atoms with fluorine and the remaining two hydrogen atoms with bromine (CF_2Br_2).

Question 35

Question: The main parameter characterizing mechanical foam agents is:

- a. Viscosity
- b. Heat resistant blanket rating
- c. Expansion rating
- d. Vapor sealing film activity

Correct Answer: Select (c)
Explanation: Mechanical foams are defined by their expansion ratio which is the final volume of foam to original foam solution volume prior to addition of air. Foams are further subdivided into: low expansion up to (20:1), medium expansion (20 to 200:1), and high expansion (200 to 1000:1).

Question 36

Question: Foam fire extinguishing agents can put out a fire in a flammable liquid spill by:

- a. Forming a blanket on the liquid and removing oxygen
- b. Interfering with the chemical chain reaction necessary for combustion reactions
- c. Cooling
- d. Only (a) and (c) above

Correct Answer: Select (d)
Explanation: Foams form a vapor seal blanket on the surface of the flammable material. The water used in foams provides cooling.

Question 37

Question: Foam fire extinguishing agents should never be used on:

- a. Class A fires
- b. Class C fires
- c. Class B fires
- d. Class A and B fires

Correct Answer: Select (b)
Explanation: Foam should never be used on class C (electrical) fires because many foams are conductors of electricity and the user may receive injury or fatal shock.

Question 38

Question: The primary mechanism by which carbon dioxide fire extinguishing agents put out a fire is by:

- a. Reducing the oxygen concentration in the atmosphere surrounding the fire.
- b. Forming a blanket and eliminating oxygen.
- c. Removing the fuel.
- d. Only (b) and (c) above.

Correct Answer: Select (a)
Explanation: CO_2 extinguishing agent dilute the atmosphere immediately surrounding the fire to concentrations below its Lower Flammability Limit (LFL).

Question 39

Question: Carbon dioxide extinguishing agents should not be used on which of the following materials on fire?

- a. Gasoline
- b. Organic peroxides
- c. Magnesium
- d. Only (b) and (c) above.

Correct Answer: Select (d)
Explanation: Carbon dioxide is not effective on fires in materials that contain their own source of oxygen such as

organic peroxides. Fires involving reactive metals such as magnesium, sodium, and potassium can not be extinguished by CO_2 because CO_2 becomes decomposed.

Question 40

Question: Considering that helium and nitrogen are both inert gases with thermal capacities of 5.0 and $7.8\text{ Btu/mole}/^\circ\text{F}$ respectively, which of the following statements is correct?

- a. Helium would be more effective than nitrogen as an extinguishing agent on a fire to provide inerting.
- b. Nitrogen would be more effective than helium in extinguishing a fire.
- c. A 50/50 mixture of nitrogen and helium would be better than either helium or nitrogen alone.
- d. Helium or nitrogen cannot extinguish a fire.

Correct Answer: Select (b)
Explanation: Inert gases such as nitrogen and helium extinguish a fire by combination of two mechanisms: by reducing the concentration of oxygen to values below the LFL, and also by reducing the temperature of the flame/vapor mixture to values below the adiabatic flame temperature. In this case both nitrogen and helium would contribute equally to the first mechanism (inerting). However, nitrogen due to its higher value of the thermal capacity contributes more than helium to reduction of the flame temperature, and therefore, is more effective in putting out a fire compared to helium.

Question 41

Question: Ordinary dry chemical fire extinguishing agents can be used on which of the following classes of fires:?

- a. Class B
- b. Class A and C fires
- c. Class B, C, and D fires
- d. Class A, B and D fires

Correct Answer: Select (a)
Explanation: "Ordinary Dry Chemical" agents are designed for use on Class B (flammable liquids, gases, and greases). On the other hand, "Multi-purpose Dry Chemicals" are suitable for use on Class A, B, and C fires.

Question 42

Question: One of the major problems associated with use of large quantities of dry chemical agents on a fire is:

- a. Production of toxic gases
- b. Breakage of foam layer causing re-ignition
- c. Interference with visibility
- d. Depletion of the protective ozone layer in the atmosphere

Correct Answer: Select (c)
Explanation: Dry chemical extinguishing agents are made of non-toxic chemicals. Use of large quantities of these agents on fire could cause temporary breathing difficulty and serious interference with visibility.

Question 43

Question: A dry powder extinguishing agent can:

- a. Put out all metal fires
- b. Also be used on Class C fires
- c. Not be designed for radioactive metal fires
- d. Put out those metal fires for which it is designed

Correct Answer: Select (d)
Explanation: Dry powder fire extinguishing agents can only be used for extinguishing fires in the metals for which they are designed.

Question 44

Question: Which of the following are examples of class D fire extinguishing agents?

- a. Graphite powder
- b. Talc
- c. Sand
- d. All of the above.

Correct Answer: Select (d)
Explanation: All have been used to smother metal fires.

Question 45

Question: All of the following extinguishing agents can be used to extinguish a Class A fire EXCEPT:

- a. Dry powder
- b. Dry chemical
- c. Carbon dioxide
- d. Halon

Correct Answer: Select (a)
Explanation: Dry powder can only be used on metal fires. Dry chemical can stop a Class A fire by both cooling and interfering with the chemical chain reactions. Halon can extinguish a Class A fire by interfering with chemical chain reaction. Carbon dioxide dilutes the mixture of fuel vapor and oxygen to values below LFL.

Question 46

Question: Which of the following are types of gas-sensing fire detector systems?

- a. Automatic and manual
- b. Semiconductor and catalytic
- c. Pressure-temperature and volume systems
- d. Ionization and photoelectric

Correct Answer: Select (b)
Explanation: The semiconductor fire-gas detectors respond to gases which have either oxidizing or reducing capabilities by creating electrical charges in a semiconductor. The catalytic type has an element which catalyzes oxidation of combustible gases and detects the resulting temperature rise associated with the oxidation reactions.

Question 47

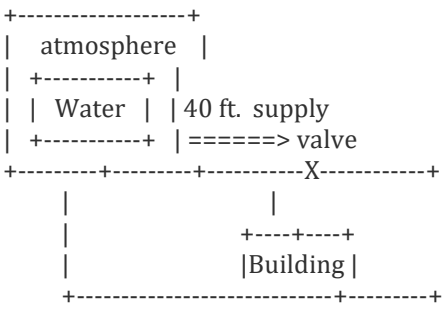
Question: The major reason behind unsatisfactory performance of sprinkler system has been:

- a. Human error
- b. Rupture in main waterline
- c. Clogged sprinkler heads
- d. Water supply tank empty

Correct Answer: Select (a)
Explanation: Approximately 35% of sprinkler failures have been caused by shutting the water supply to them during a fire.

Question 48

Question: Water is delivered by gravity to a sprinkler main supply line as shown in the diagram. When the supply valve is closed, the gauge pressure in the main supply line is: (density of water $= 62.4\text{ lbm/ft}^3$)



- a. 17.3 psig
- b. 2,400.0 psig
- c. 123.5 psig
- d. 628.0 psig

Correct Answer: Select (a)
Explanation:
$$P = \rho \frac{g}{g_c} h$$

Where:
P is pressure
g is acceleration of gravity (32.2 ft/sec^2)
g_c is a factor equal to 1 in metric and 32.2 in American Engineering system
h is height of liquid

$$P = 62.4 \times \left(\frac{32.2}{32.2}\right) \times 40 = 2,496\text{ lbf/ft}^2$$
$$P = \frac{2,496}{144} = 17.33\text{ psig}$$

Question 49

Question: Which of the following diagrams demonstrates the operation of a regular dry pipe sprinkler system?

- a. Water Tank -> Valve (closed) -> Sprinkler Pipe (unpressurized)
- b. Water Tank (P_2) -> Valve (closed) -> Sprinkler Pipe (P_1, air under pressure, $P_1 > P_2$)
- c. Nitrogen Tank -> Valve -> Sprinkler Pipe
- d. Water Tank -> Valve -> Sprinkler Pipe (air under pressure)

Correct Answer: Select (b)
Explanation: Regular dry pipe automatic sprinkler systems are connected to a supply of air or nitrogen under pressure. Heat from a fire causes a valve to open and reduce the pressure on air or nitrogen supply to a value that water pressure on the supply side of the system causes the valve to open and establish flow of water to the sprinkler heads.

Question 50

Question: Which of the following statement(s) is (are) true regarding deluge type automatic sprinkler systems?

- a. Sprinklers are opened in response to pressure reduction in air or nitrogen tank.
- b. Use 15% carbon dioxide to deluge the protected area.
- c. All sprinkler heads are open at all times.
- d. Only (a) and (b) above.

Correct Answer: Select (c)
Explanation: In "deluge" type sprinkler systems the sprinkler heads are open at all times. The deluge valve responds to a fire detection system. When the valve is opened the water flows into and out of the sprinklers deluging the protected area.

Question 51

Question: Fire extinguishment by water spray is accomplished by:

- a. Cooling and smothering
- b. Emulsification
- c. Dilution
- d. All of the above

Correct Answer: Select (d)
Explanation: Water spray provides cooling and smothering by steam. Emulsification and dilution are also contributing factors.

Question 52

Question: The total pressure loss due to friction in 1,000 ft. of a pipe which has an inside diameter of 8 inches and a coefficient of friction of 150 with a flow rate of 1,000 gpm is approximately:

- a. 10 psi
- b. 6 psi
- c. 15 psi
- d. 8 psi

Correct Answer: Select (b)
Explanation: The formula for calculation of frictional losses in a piping system is:

$$P_d = \frac{4.52 Q^{1.85}}{C^{1.85} d^{4.87}}$$

where:
P_d is pressure loss due to friction in psi/ft.
Q is flow rate in gpm
d is pipe inside diameter in inches
C is coefficient of friction

For the problem at hand:
$$P_d = \frac{4.52 (1000)^{1.85}}{(150)^{1.85} (8)^{4.87}} = 6.04 \times 10^{-3}\text{ psi/ft.}$$

For 1,000 ft. of piping, the total loss is:
$$1,000 \times 6.04 \times 10^{-3} = 6.04\text{ psi}$$

Question 53

Question: An automatic sprinkler system has an orifice discharge coefficient of 4.2. At a system pressure of 60 psi, the amount of water delivery is:

- a. 60.8 gpm
- b. 42.5 gpm
- c. 32.5 gpm
- d. 53.6 gpm

Correct Answer: Select (c)
Explanation: The formula for orifice flow is:

$$P = \left(\frac{Q}{K}\right)^2$$

where:
P is pressure expressed in units of psi
K is orifice discharge coefficient
Q is rate of flow expressed in gallons per minute (gpm)

For the problem at hand:
$K = 4.2$
$P = 60\text{ psi}$

$$60 = \left(\frac{Q}{4.2}\right)^2$$
$$Q = \sqrt{60} \times 4.2 = 7.746 \times 4.2 = 32.53\text{ gpm} \approx 32.5\text{ gpm}$$

Question 54

Question: The K-factor for an orifice in a sprinkler system which delivers 60 gpm at 80 psi is:

- a. 5.2
- b. 4.8
- c. 7.6
- d. 6.7

Correct Answer: Select (d)
Explanation:
$$P = \left(\frac{Q}{K}\right)^2$$
$$80 = \left(\frac{60}{K}\right)^2$$
$$80 K^2 = 3,600$$
$$K^2 = 45$$
$$K = \sqrt{45} = 6.708 \approx 6.7$$

Question 55

Question: An orifice in a sprinkler system has a K-factor of 4.9. When water delivery through the head is 60 gpm, the water pressure is equal to:

- a. 120.5 psi
- b. 490.3 psi
- c. 149.9 psi
- d. 68.3 psi

Correct Answer: Select (c)
Explanation:
$$P = \left(\frac{Q}{K}\right)^2 = \left(\frac{60}{4.9}\right)^2 = (12.245)^2 = 149.94\text{ psi}$$

Question 56

Question: Water with a density of $62\text{ lb per cubic foot}$ is flowing through an 8 inch diameter horizontal piping system. The head loss across a 1,000 ft. section of this piping system is 70 ft. The residual pressure at point A is 50 psi and the velocity at point A is 6 ft/sec. If the velocity at point B is 45 ft/sec, the value of residual pressure at point B is:

- a. 60.3 psi
- b. 100.5 psi
- c. 6.5 psi
- d. 90.4 psi

Correct Answer: Select (c)
Explanation: This problem is solved by Bernoulli's Theorem:

$$\frac{V_A^2}{2g} + \frac{P_A}{\rho} + Z_A = \frac{V_B^2}{2g} + \frac{P_B}{\rho} + Z_B + h_L$$

where:
V is velocity
P is residual pressure
ρ is density
Z is elevation
h_L is head loss
g is acceleration of gravity (32.2 ft/sec^2)

For the problem at hand:
$$\frac{6^2}{2 \times 32.2} + \left(\frac{50\text{ psi}}{62\text{ lb/ft}^3}\right) \left(\frac{144\text{ in}^2}{1\text{ ft}^2}\right) + 0 = \frac{45^2}{2 \times 32.2} + \frac{P_B}{62} + 0 + 70$$

$$0.559 + 116.129 = 31.444 + \frac{P_B}{62} + 70$$
$$116.688 = 101.444 + \frac{P_B}{62}$$
$$\frac{P_B}{62} = 15.244 \implies P_B = 945.13\text{ psf}$$
$$P_B = \frac{945.13}{144} = 6.56\text{ psi} \approx 6.5\text{ psi}$$

Question 57

Question: The design of a sprinkler system requires 2,000 gpm at 50 psi. The gpm requirement at 60 psi is:

- a. 1,560 gpm
- b. 2,875 gpm
- c. 1,475 gpm
- d. 2,190 gpm

Correct Answer: Select (d)
Explanation: The following relationship applies:

$$\frac{Q_1}{Q_2} = \frac{\sqrt{P_1}}{\sqrt{P_2}}$$

where:
Q is gallons per minute (gpm)
P is pressure in psi

For the problem at hand:
$Q_1 = 2,000\text{ gpm}$, $P_1 = 50\text{ psi}$, $P_2 = 60\text{ psi}$

$$\frac{2000}{Q_2} = \frac{\sqrt{50}}{\sqrt{60}} = \sqrt{\frac{5}{6}} = 0.91287$$
$$Q_2 = \frac{2000}{0.91287} = 2,190.89\text{ gpm} \approx 2,190\text{ gpm}$$

Question 58

Question: If the pressure at the top of the riser in a sprinkler system is increased by 50%, the increase in the gpm requirement will increase by:

- a. 22%
- b. 25%
- c. 50%
- d. 100%

Correct Answer: Select (a)
Explanation:
$$\frac{P_2}{P_1} = 1.5 \implies \frac{P_1}{P_2} = 0.666$$
$$\frac{Q_1}{Q_2} = \sqrt{0.666} = 0.816 \implies \frac{Q_2}{Q_1} = 1.2247$$

This means that the gpm requirements will increase by approximately 22 percent when pressure has increased by 50 percent.

Question 59

Question: Water is flowing through a 10 inch inside diameter pipe at a rate equal to 2,000 gpm. The velocity pressure in the pipe is equal to:

- a. 0.448 psi
- b. 0.200 psi
- c. 0.753 psi
- d. 0.176 psi

Correct Answer: Select (a)
Explanation: The relationship between velocity pressure, pipe diameter, and flow rate is:

$$P_v = \frac{Q^2}{891 d^4}$$

where:
P_v = velocity pressure in psi
Q = flow rate in gpm
d = pipe diameter in inches

For the problem at hand:
$d = 10\text{ inches}$, $Q = 2,000\text{ gpm}$

$$P_v = \frac{(2000)^2}{891 \times (10)^4} = \frac{4,000,000}{8,910,000} = 0.4489\text{ psi} \approx 0.448\text{ psi}$$

Question 60

Question: If the velocity pressure in a 6 inch diameter pipe is equal to 0.630 psi the flow rate through the pipe is approximately equal to:

- a. 360 gpm
- b. 853 gpm
- c. 562 gpm
- d. 749 gpm

Correct Answer: Select (b)
Explanation:
$$P_v = \frac{Q^2}{891 d^4}$$
$$Q^2 = 891 P_v d^4 \implies Q = \sqrt{891 P_v d^4}$$
$$Q = \sqrt{891 \times 0.630 \times 6^4} = \sqrt{891 \times 0.630 \times 1296} = \sqrt{727,485.84} = 852.92\text{ gpm} \approx 853\text{ gpm}$$

Question 61

Question: The velocity pressure in a pipe is equal to 0.5 psi. If 1,000 gpm is flowing through this pipe, the pipe diameter is approximately equal to:

- a. 14 inches

- b. 21 inches
- c. 7 inches
- d. 4 inches

Correct Answer: Select (c)
Explanation:
$$P_v = \frac{Q^2}{891 d^4}$$
$$d^4 = \frac{Q^2}{891 P_v} = \frac{(1000)^2}{891 \times 0.5} = \frac{1,000,000}{445.5} = 2,244.67$$
$$d^2 = \sqrt{2,244.67} = 47.378$$
$$d = \sqrt{47.378} = 6.88\text{ inches} \approx 7\text{ inches}$$

Question 62

Question: Which of the following is the correct classification for flammable liquids?

- a. Class 1A, 2B, 3C
- b. Class 1A, 1B, 1C
- c. Class 1A, 2B
- d. Class 1A

Correct Answer: Select (b)
Explanation:
Class IA: Flash point below 73°F; boiling point below 100°F
Class IB: Flash point below 73°F; boiling point at or above 100°F
Class IC: Flash point at or above 73°F, but below 100°F
Combustible liquids are also classified into three groups: Class II, IIIA, IIIB.

Question 63

Question: The maximum amount of Class 1A flammable liquid that can be stored in a laboratory unit with a sprinkler system is:

- a. 2,000 gallons
- b. 500 gallons
- c. 1,500 gallons
- d. 1,200 gallons

Correct Answer: Select (d)
Explanation: Only NFPA 45-approved amounts of flammable liquids shall be stored in laboratory units outside of flammable liquid storage rooms.

Question 64

Question: Not more than _____ gallons of Class I, Class II, and Class IIIA liquids shall be stored in an approved flammable storage cabinet.

- a. 40
- b. 50
- c. 120

- d. 100

Correct Answer: Select (c)
Explanation: The correct answer is 120 gallons. Also, no more than 60 gallons of the total shall be Class I and Class II liquids.

Question 65

Question: All of the following are true in regards to a jockey pump EXCEPT:

- a. A jockey pump is a small pump used to keep a pressure on a fire main.
- b. The jockey pump keeps a high positive pressure on the discharge side of fire pumps.
- c. It is a spare pump used to provide cooling for highly-exothermic chemical reactions.
- d. If you put packing in the stuffing box of a fire pump, the pump will run almost continually to compensate for the packing leakage.

Correct Answer: Select (c)
Explanation: Statements (a), (b), and (d) are all true in regards to jockey pumps. The purpose of a jockey pump is to maintain water pressure in the pipes in a predetermined range when there is no fire. This helps the main fire pump to not run all the time and also prevents the sprinklers go off randomly.

Question 66

Question: A fire water supply system delivers 8,000 gallons per minute (gpm) when the pressure drops from 85 psig to 40 psig. What is the flow rate if the pressure drops from 85 psig to 25 psig?

- a. 8,700 gpm
- b. 5,200 gpm
- c. 9,344 gpm
- d. 16,000 gpm

Correct Answer: Select (c)
Explanation: The formula to solve this problem is:

$$Q_2 = Q_1 \left[\frac{(S - R_2)^{0.54}}{(S - R_1)^{0.54}} \right]$$

Where:
Q_2 is the final flow rate
Q_1 is the initial flow rate
S is the initial pressure
R_1 is the initial residual pressure
R_2 is the final residual pressure

For the problem at hand:
$$Q_2 = 8000 \left[\frac{(85 - 25)^{0.54}}{(85 - 40)^{0.54}} \right] = 8000 \left[\frac{60^{0.54}}{45^{0.54}} \right] = 8000 \left(\frac{9.13}{7.82} \right) = 9,344.52\text{ gpm}$$

Question 67

Question: Sodium azide solution can result in serious explosion hazards if the waste is disposed in a sink made of _____.

- a. Brass
- b. Lead
- c. Copper
- d. All of the above

Correct Answer: Select (d)
Explanation: Sodium azide solution in contact with brass, lead, or copper can cause serious explosion hazards.

Question 68

Question: Fire detection and alarm systems use distinctly different signals that are automatically transmitted to an approved central station, remote supervising station, or proprietary supervising station. Which of the following is NOT a signal used in these systems?

- a. Supervisory
- b. Electromagnetic
- c. Alarm
- d. Trouble

Correct Answer: Select (b)
Explanation: The three types of signals used in fire detection are supervisory, alarm, and trouble.

Question 69

Question: Which of the following is NOT a mode of heat transfer?

- a. Conduction
- b. Convection
- c. Combustion
- d. Radiation

Correct Answer: Select (c)
Explanation: Conduction, convection, and radiation are the three modes of heat transfer. Combustion is a chemical reaction.

Question 70

Question: Hazardous locations are classified by the National Electric Code (NEC) as Class I, Class II, and Class III locations. Class III locations are those in which there may be:

- a. Flammable vapors
- b. Flammable gases
- c. Flammable liquids
- d. Ignitable fibers

Correct Answer: Select (d)
Explanation: Class III locations are those in which there may be ignitable fibers.

Question 71

Question: Which of the following color coding is used for a sprinkler head designed for a maximum ceiling temperature of 300°F (149°C)?

- a. White
- b. Green
- c. Red
- d. Orange

Correct Answer: Select (c)
Explanation: Blue ($107\text{--}225^\circ\text{F}$), Red ($149\text{--}300^\circ\text{F}$), Green ($191\text{--}375^\circ\text{F}$), Orange ($246\text{--}475^\circ\text{F}, 329\text{--}625^\circ\text{F}$). Red is used for $300^\circ\text{F} / 149^\circ\text{C}$.

Question 72

Question: Which of the following is NOT true in regard to Halon fire extinguishing agents?

- a. Halons are made by halogenating a hydrocarbon.
- b. Halons stop the spread of fire by chemically disrupting combustion.
- c. The only disadvantage of Halons is that they leave a residue on the equipment after putting out the fire.
- d. Halon is rated for Class B and Class C fires, but is also effective on Class A fires.

Correct Answer: Select (c)
Explanation: Halons are made by halogenating hydrocarbons. Although hydrocarbons are flammable, the halogenated hydrocarbons lose their flammability characteristic as they become more halogenated. Halons are especially effective for electrical fires in sensitive electronic equipment, because they leave no residue and therefore do not damage the equipment.

Question 73

Question: The term boiling liquid expanding vapor explosion (BLEVE) refers to which of the following?

- a. An explosion caused by a flammable liquid at its boiling point
- b. The failure of a container holding a liquid above its boiling point, which may result in an explosion
- c. The volume of vapor expanding over the surface of a flammable liquid, which may result in an explosion
- d. All of the above

Correct Answer: Select (b)
Explanation: BLEVE is best described by statement (b). The consequences of BLEVE include overpressure blast wave, fireball, and vessel fragment hazards.

Question 74

Question: Which of the following statements is true in regard to ionization smoke detectors?

- a. They have a small amount of radioactive material between two electrically charged plates.
- b. They ionize the air and cause current to flow between the plates.
- c. Smoke disrupts the flow of ions, thus reducing the flow of current, which in turn activates the alarm.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Another type of smoke detector is photoelectric. This type aims a light source into a sensing chamber at an angle away from the sensor. Smoke enters the chamber, reflecting light onto the light sensor, triggering the alarm.

Question 75

Question: Which of the following statements BEST describes the term hypergolic?

- a. The ignition of a fuel when it comes into contact with an oxidizer without the presence of an outside source of ignition
- b. A flammable liquid that can create a hyperbola-shaped fireball when on fire.
- c. An above-standard goal-oriented safety, health, and environmental program
- d. The angle of incidence of a hyperbola-shaped dome in construction safety

Correct Answer: Select (a)
Explanation: Oxidizers, such as hydrogen peroxide, have excess oxygen in their molecules and can release this excess oxygen. If the reaction between an oxidizer and fuel can generate enough heat to raise the mixture temperature to above its auto-ignition temperature, no outside source of ignition is needed to cause a fire. Flammables must be separated from oxidizers.

Question 76

Question: Which of the following is NOT a pyrophoric material?

- a. Metal hydrides
- b. White phosphorus
- c. Alkyl lithiums
- d. Iron oxides

Correct Answer: Select (d)
Explanation: Pyrophoric materials are substances that ignite instantly upon exposure to oxygen. No source of ignition is needed. They can also be water-reactive, which produces heat and hydrogen.

Question 77

Question: The first step in responding to spills of hazardous material should be to:

- a. Contain the spill
- b. Protect people
- c. Identify the substance
- d. Read SDS

Correct Answer: Select (b)
Explanation: Although identifying the substance determines the most effective clean up procedure and protective equipment needed, the first step is to ensure that people are protected. This includes both responders and other people.

Question 78

Question: Employees who use portable fire extinguishers can often put out small fires or control a fire until additional help arrives. Before an emergency occurs, employers must decide whether employees are authorized to use fire extinguishers or must immediately evacuate (29 CFR 1910.38). Which of the following statements is NOT true in regard to portable fire extinguishers intended for emergency response?

- a. Selection and distribution of portable fire extinguishers must be based on the class, size, and degree of workplace fire hazards.
- b. Mounting, location, and identification of the extinguishers must be done in such a way so they are readily accessible in an emergency and will not subject employees to potential injury.
- c. Ensure that the travel distance from employee to the nearest extinguisher is 100 feet or less.
- d. Maintain fire extinguishers. Maintenance includes monthly visual inspections, hydrostatic testing, annual internal examinations, and all associated documentation.

Correct Answer: Select (c)
Explanation: The Occupational Safety and Health Administration require the following for portable fire extinguisher to be used in an emergency (29 CFR 1910.157):

- Select and distribute portable fire extinguishers based on the class, size, and degree of workplace fire hazards.
- Mount, locate, and identify the extinguishers so they are readily accessible in an emergency and will not subject employees to potential injury.
- Provide only approved portable fire extinguishers.
- Maintain fire extinguishers. Maintenance includes monthly visual inspections, hydrostatic testing, annual internal examinations, and all associated documentation.
- Ensure that the travel distance from employee to the nearest extinguisher is appropriate for the fire class. The travel distance for class A extinguishers cannot exceed 75 feet. The maximum travel distance for class B is 50 feet and there is no maximum travel distance specified for class C extinguishers.

Question 79

Question: In an Emergency, employee alarm systems alert employees to begin implementing emergency action. All of the following are true in regard to alarm systems EXCEPT:

- a. Provide a distinctive and perceivable alarm system for emergency action.
- b. Ensure that all equipment used for alarm systems is approved.
- c. Ensure that all equipment used for alarm systems has spare components.
- d. Test alarms once a week. Follow special safety requirements for testing.

Correct Answer: Select (d)
Explanation: Employee alarm systems alert employees to begin implementing emergency action. OSHA requires the following:

- Provide a distinctive and perceivable alarm system for emergency action or safe evacuation.
- Ensure that all equipment used for alarm systems is approved and spare components are available.
- Test alarms at the frequency required. Follow special safety requirements for testing or restoring alarms.

Question 80

Question: To prepare for any contingency, an emergency action plan establishes procedures that prevent fatalities, injuries, and property damage. Possible emergency scenarios based on the nature of the workplace and its surroundings must be identified. A written emergency action plan must also be prepared. The plan must include all of the following EXCEPT:

- a. The fire and emergency reporting procedures
- b. Procedures for emergency evacuation, including the type of evacuation and exit routes.
- c. Procedures for those who remain to operate critical operations prior to Evacuation.
- d. The name, telephone number, and other relevant information of outside emergency personnel who are expected to respond to onsite emergencies.

Correct Answer: Select (d)
Explanation: OSHA requires the employer to:

- Identify possible emergency scenarios based on the nature of the workplace and its surroundings.
- Prepare a written emergency action plan. The plan does not need to be written and may be communicated orally if there are 10 or fewer employees. At a minimum, the plan must include: The fire and emergency reporting procedures; Procedures for emergency evacuation, including the type of evacuation and exit routes; Procedures for those who remain to operate critical operations prior to evacuation; Procedures to account for employees after evacuation; Procedures for employees performing rescue and medical duties; and Names of those to contact for further information or explanation about the plan.

Question 81

Question: Process safety management (PSM) of highly hazardous chemicals focuses on preventing or minimizing consequences from a catastrophic release of toxic, reactive, flammable, or explosive chemicals. Successful PSM emergency planning relies on implementing requirements from 29 CFR 1910.38 and/or 1910.120(q). All of the following are elements of a sound PSM emergency plan EXCEPT:

- a. Conduct a reliability analysis for each component of covered processes, and update and revalidate the analysis every 2 years.
- b. Incorporate emergency shutdown actions and operations into the written operating procedures for each process.
- c. Implement an emergency action plan for the facility as described in 29 CFR 1910.38.
- d. Maintain the mechanical integrity of PSM emergency systems and alarms.

Correct Answer: Select (a)
Explanation: OSHA's Process Safety Management Standard (29 CFR 1910.119) requires the employer to:

- Conduct a Process Hazard Analysis (PHA) for each covered process, and update and revalidate the PHA every 5 years.
- Incorporate emergency shutdown actions and operations into the written operating procedures for each process. Include conditions that require emergency action and the qualified operator responsible for performing these procedures.
- Implement an emergency action plan for the facility as described in 29 CFR 1910.38.
- Maintain the mechanical integrity of PSM emergency systems and alarms.
- If employees are expected to handle an emergency release rather than promptly evacuate, implement an emergency response plan according to 1910.120(q).
- Provide proper response and personal protective equipment for emergency responders under the plan.

Question 82

Question: Emergency planning and response are required safety and health program elements that help minimize employee exposure and injury in hazardous waste sites. All of the following are elements of a sound emergency response plan for hazardous waste site emergencies EXCEPT:

- a. If employees are to immediately evacuate in an emergency, develop an emergency action plan in accordance with 29 CFR 1910.38.
- b. Identify emergency response personnel and responsibilities in the site program's organizational structure.
- c. Identify the name and telephone numbers of federal, state, and local emergency response personnel.
- d. Identify personal protective equipment (PPE) and other equipment for emergency response in the emergency response plan.

Correct Answer: Select (c)
Explanation: OSHA requires the employer to do the following:

- If employees are to immediately evacuate in an emergency, develop an emergency action plan in accordance with 29 CFR 1910.38.
- If employees may assist in handling the emergency, then the following requirements apply: As a separate section of the site safety and health plan, develop a written emergency response plan. The plan must be implemented before site operations begin and should be integrated with those of other local, state, and Federal agencies. Identify emergency response personnel and responsibilities in the site program's organizational structure. Include emergency response training details in the comprehensive site work plan. Identify personal protective equipment (PPE) and other equipment for emergency response in the emergency response plan. Install alarm systems that meet requirements of 29 CFR 1910.165. Share site-specific emergency response procedures with contractors and subcontractors. Under certain circumstances, emergency response personnel may qualify for the medical surveillance program.

Question 83

Question: In responding to hazardous substance emergencies, if the employer has decided that employees may assist in handling the emergency, a written emergency response plan must be developed and implemented. This plan must include all of the following EXCEPT:

- a. Pre-emergency planning and coordination
- b. Personnel roles, lines of authority, training, and communication
- c. Emergency recognition and prevention
- d. Exclude employees with disability from taking part in emergency operations

Correct Answer: Select (d)
Explanation: Depending on the type of emergency and disability, some employees with disability may be able to help in handling an emergency.

Question 84

Question: When an employer establishes a fire brigade to respond to workplace fires, it must meet organizational, training, and personal protective equipment requirements. All of the following are procedural, program, and/or equipment requirements that the employer must meet EXCEPT:

- a. Prepare and maintain a fire brigade written organizational statement and document fire brigade member training information in the organizational statement.
- b. Ensure employees expected to fight fires are physically capable to perform assigned duties.
- c. Inspect and maintain fire-fighting equipment monthly. Inspect respirators and fire extinguishers annually. Remove and replace damaged equipment.
- d. Supply protective clothing with components to protect the head, body, and extremities at no cost to the employees.

Correct Answer: Select (c)
Explanation: The employer must do the following:

- Prepare and maintain a fire brigade written organizational statement. Document fire brigade member training information in the organizational statement.
- Ensure employees expected to fight fires are physically capable to perform assigned duties.
- Inspect and maintain fire-fighting equipment annually. Inspect respirators and fire extinguishers monthly.
- Supply protective clothing with components to protect the head, body, and extremities at no cost to the employee.
- Ensure fire brigade members are provided with and use compliant respirators.

Question 85

Question: All of the following are true in regard to emergency respirators EXCEPT:

- a. Clean and disinfect emergency respirators at least once every six months.
- b. Select and provide respirators suitable for the intended purpose.
- c. When respirators are required, establish and implement a written respiratory protection program that includes procedures for proper respirator use in emergency.
- d. Ensure that emergency respirators are properly stored and inspected.

Correct Answer: Select (a)
Explanation: OSHA requires the employer to do the following:

- Select and provide respirators suitable for the intended purpose.
- When respirators are required, establish and implement a written respiratory protection program that includes procedures for proper respirator use in emergency situations. Update the program when changes in workplace conditions affect respirator use.
- Clean and disinfect emergency respirators after each use. Ensure that emergency respirators are properly stored and inspected. Certify emergency-only respirators by a certification tag or other proper documentation method.
- For entry into Immediately Dangerous to Life and Health (IDLH) environments, provide retrieval equipment or equivalent means for rescue of employees. Maintain required personnel and communications for emergency rescue.
- For interior structural firefighting, ensure that at least two employees who remain in contact with one another enter the structure.

Question 86

Question: Which of the following is true in regards to workplace violence?

- a. Workplace violence is any act or threat of physical violence, harassment, intimidation, or other threatening disruptive behavior.
- b. It ranges from threats and verbal abuse to physical assaults and even homicide.

- c. In most workplaces where risk factors can be identified, the risk of assault can be prevented or minimized.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Workplace violence is any act or threat of physical violence, harassment, intimidation, or other threatening disruptive behavior that occurs at the work site. It ranges from threats and verbal abuse to physical assaults and even homicide. It can affect and involve employees, clients, customers and visitors.

In most workplaces where risk factors can be identified, the risk of assault can be prevented or minimized if employers take appropriate precautions. One of the best protections employers can offer their workers is to establish a zero-tolerance policy toward workplace violence. This policy should cover all workers, patients, clients, visitors, contractors, and anyone else who may come in contact with company personnel.

By assessing their worksites, employers can identify methods for reducing the likelihood of incidents occurring. OSHA believes that a well written and implemented Workplace Violence Prevention Program, combined with engineering controls, administrative controls and training can reduce the incidence of workplace violence in both the private sector and Federal workplaces.

Question 87

Question: A comprehensive crime prevention assessment should ask all of the following EXCEPT:

- a. What is the prevailing attitude toward security?
- b. Who is responsible for the overall security program?
- c. How are security policies enforced?
- d. What kind of physical safety systems are presently used in other organizations?

Correct Answer: Select (d)
Explanation: Other pertinent questions include:

- What is your target potential?
- What resources are available locally and how rapid are the response times for fire, police and ambulance?
- What kind of physical security systems and controls are presently used?
- Do the available security resources, policies and procedures meet the potential threat?

Question 88

Question: All of the following are guidelines that can increase security except:

- a. Install key-card access systems at main entrances and on other appropriate doors.
- b. Issue access control badges, with recent photographs, to all employees and authorized contractors.
- c. Upgrade perimeter control systems with intercoms and closed circuit monitoring devices.
- d. Provide master and extra keys to all employees

Correct Answer: Select (d)
Explanation: The master and extra keys should be locked in the security office. Other guidelines that can increase security include the following:

- Issue access control badges, with recent photographs, to all employees and authorized contractors.
- Upgrade perimeter control systems with intercoms and closed circuit monitoring devices.
- Keep master and extra keys locked in a security office.

Question 89

Question: According to OSHA regulations, companies with _____ or more employees must have a written emergency action plan.

- a. 5
- b. 10
- c. 1
- d. 20

Correct Answer: Select (b)
Explanation: OSHA requires that companies with 10 or more employees develop a written emergency action plan. Smaller companies may communicate their emergency action plan to their employees orally.

Question 90

Question: Emergency procedures, including the handling of any toxic chemicals, should include all of the following EXCEPT:

- a. Escape procedures and escape route assignments.
- b. Special procedures for employees who perform or shut down critical plant operations.
- c. Systems to account for all employees after evacuation and for information about the plan.
- d. Procedures for segregation of toxic chemicals based on their toxicity level.

Correct Answer: Select (d)
Explanation: Emergency procedures must also address rescue and medical duties for employees who perform them as well as means for reporting fires and other emergencies.

Question 91

Question: OSHA requires that the employer designates an emergency response coordinator and a backup coordinator. Duties of the coordinator and employer include all of the following EXCEPT:

- a. Determining what emergencies may occur and seeing that emergency procedures are developed to address each situation.
- b. Directing all emergency activities including evacuation of personnel.
- c. Ensuring that outside emergency services are notified when necessary.
- d. Ensuring that a member of the local fire department and police serve on the emergency committee.

Correct Answer: Select (d)
Explanation: The emergency response coordinator must also be able to direct the shutdown of plant operations when necessary.

Question 92

Question: You are a supervisor in the manufacturing department of a company. You observe an employee ignoring the department's safety rules. Which of the following is the most effective disciplinary action against the employee?

- a. Immediately inform the HR department in writing
- b. Suspend the employee and provide training
- c. Provide the employee with a written warning
- d. Use verbal reprimand first, and if the behavior continues, issue a written warning followed by suspension and eventual termination.

Correct Answer: Select (d)
Explanation: This would be the best course of action to follow.

Question 93

Question: A rating of four in the blue section of NFPA 704 system indicates:

- a. Highly flammable material.
- b. Extremely high health hazard.
- c. Moderate reactivity.
- d. Highly ignitable oxidizer.

Correct Answer: Select (b)
Explanation: In NFPA 704 system, red indicates flammability, blue indicates health hazards, yellow indicates reactivity, and the color white is reserved for communicating special information such as a material being an oxidizer or water reactive. In each section, there is a rating between zero and four with zero indicating no hazards and four indicating extreme hazards.

Question 94

Question: In fire-fighting operations, water can extinguish a fire by which of the following mechanisms?

- a. Blocking oxygen
- b. Stopping chemical chain reactions
- c. Cooling
- d. All of the above

Correct Answer: Select (c)
Explanation: Water has an excellent evaporative cooling effect on a fire. Water should never be used on certain class B (flammable liquids), class D (metal fires), and class C (electrical) fires. If a flammable liquid is insoluble in water and also lighter than water, it would float on top of the water and spread the fire. Use of water on electrical fires could pose a fatal shock hazard.

Question 95

Question: The most important step in prevention of workplace violence is:

- a. A clear non-harassment policy by the management.
- b. Training.
- c. Random drug and alcohol testing.
- d. Isolation of employees who have a tendency to harass others.

Correct Answer: Select (a)
Explanation: Management commitment and a clear non-harassment policy is the most important step in prevention of workplace violence.

Question 96

Question: Which of the following is the BEST tool for minimizing loss of life during a fire?

- a. Compatible storage of flammables.
- b. Use of fire resistant materials.
- c. Fire proofing of structural systems
- d. Automatic sprinkler systems.

Correct Answer: Select (d)
Explanation: Use of automatic sprinkler systems has proven to minimize loss of life during fires.

Question 97

Question: When transferring flammable or combustible liquids between containers, which of the following safety measures must be undertaken?

- a. Provide electrical grounding for the larger container.
- b. Ground both containers.
- c. Bond the containers with an electrical conductor.
- d. The only safety measure to follow for this situation is elimination of all sources of ignition.

Correct Answer: Select (c)
Explanation: Bonding the containers together does not eliminate the buildup of static charges. However, it makes the potential difference between the containers equal so there is no driving force for the flow of static charges which could act as a source of ignition.

Question 98

Question: Which of the following represents the BEST classification for welding and cutting operations?

- a. Friction producing work
- b. Heat producing work
- c. Hot work
- d. Mechanical spark producing work

Correct Answer: Select (c)
Explanation: Hot work is any work that involves burning, welding, using fire, or spark-producing tools, or that produces a source of ignition. Examples of hot work include welding and cutting operations.

Question 99

Question: Which of the following organizations mandates lockout/tagout programs for service or maintenance of certain types of equipment?

- a. NIOSH
- b. OSHA
- c. FEMA
- d. CPSC

Correct Answer: Select (b)
Explanation: The correct answer is OSHA.